UNIT TEST REPORT - Traffic Management System (Simulation)
Date: 03-11-2026
Environment: Ubuntu 24.04, g++ 13.3.0, C++17
Method: each tests/test_*.cpp compiled with all src/*.cpp except main.cpp and run
Result: 13 test executables run (12 unit test programs + test_stress_100), 0 failures

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=== test_console_actions ===
All console action tests passed.
=== test_console_files ===
All console file tests passed.
=== test_console_interface ===
All ConsoleInterface tests passed.
=== test_file_manager ===
All FileManager tests passed.
=== test_input_validator ===
All InputValidator tests passed.
=== test_input_validator_double ===
All double InputValidator tests passed.
=== test_managers ===
=== TC-VM: Vehicle Tests ===
PASS: Vehicle IDs are unique
PASS: First vehicle entry tick
PASS: Second vehicle entry tick
PASS: Vehicle approach is stored

=== TC-Q: Queue Tests ===
PASS: North queue size
PASS: South queue size
PASS: Total waiting count
PASS: Dequeue from non-empty queue
PASS: FIFO order
PASS: Empty queue returns false
PASS: Queue reset clears vehicles

=== TC-SIG: Signal Tests ===
PASS: TC-SIG-01: Approach states at tick 0 (N+S green, E+W red)
PASS: TC-SIG-02: North-South yellow after green duration
PASS: TC-SIG-02: East-West green after North-South yellow
PASS: TC-SIG-02: East-West yellow after green duration
PASS: TC-SIG-02: Cycle returns to North-South
PASS: TC-SIG-03: Longer green keeps North green at tick 2
PASS: TC-SIG-03: North yellow after new green duration
PASS: TC-SIG-03: East green after new yellow duration
PASS: TC-SIG-03: Invalid durations are ignored
PASS: TC-SIG-04: Manual phase applied and automatic cycle paused
PASS: TC-SIG-04: Automatic mode resumes after manual mode
PASS: TC-SAFE-01: No conflicting signals over 200 ticks and manual mode

Passed: 23 | Failed: 0
=== test_queue_manager ===
All 5 queue tests passed.
=== test_simulation_engine ===
All SimulationEngine tests passed.
=== test_statistics_manager ===
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[Statistics] Warning: negative wait time for vehicle (entryTick=10,
exitTick=5). Clamping to 0.
All StatisticsManager tests passed.
=== test_statistics_summary ===
PASS: printSummary prints output for empty statistics
PASS: no vehicles processed yet
PASS: average wait is 0 with no vehicles
PASS: two vehicles recorded
PASS: total wait time is 10
PASS: average wait is 5.0
PASS: throughput is non-negative
PASS: throughput with 0 ticks is safe
PASS: printSummary prints output with data
PASS: reset clears the processed count
All statistics summary tests passed.
=== test_stress_100 ===
PASS: 100 vehicles queued
  drained in 70 ticks, 0.077209 ms
PASS: all 100 vehicles processed
PASS: all queues empty
PASS: 100 unique vehicle IDs
PASS: no conflicting greens during run
PASS: finished within tick budget
PASS: run time within limit
PASS: average wait is valid
PASS: auto-generation produced 100 vehicles
PASS: auto-generated vehicles all processed
PASS: vehicles still queued after Stop
PASS: Reset clears queues
PASS: Reset clears statistics
PASS: Reset clears clock
PASS: 50 load/reset cycles survived
All stress tests passed.
=== test_vehicle_generation ===
Testing Vehicle Generation

PASS: Probability 0.0 generates no vehicle
PASS: Probability 1.0 always generates a vehicle
PASS: Probability 0.1 generated rate = 0.117
PASS: Invalid probabilities are rejected
PASS: Vehicle IDs are unique and increasing
PASS: Vehicle is created with correct approach and entry tick

All vehicle generation tests passed.
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